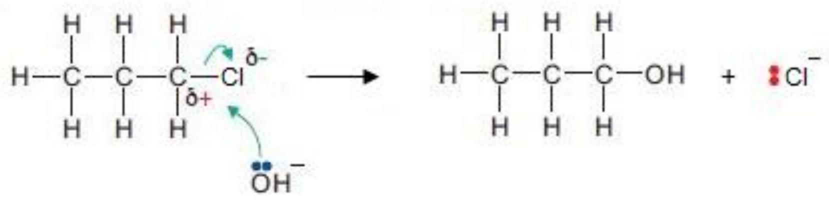
Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
12.	(a)	(i)		add aqueous NaOH and then (neutralise with) HNO_3 (1) add aqueous $AgNO_3$ (1) yellow precipitate for iodine, cream for bromine and white for chlorine (1)	3			3		3
		(ii)		**Y** is C_4H_9Cl / chlorobutane (1) award (1) for any explanation supported by a simple calculation e.g. showing that chlorine accounts for 38% of mass of chlorobutane showing that bromine (and iodine) account for greater than 40% of mass of their respective halogenoalkanes showing that the hydrocarbon chain has a mass of 57 and only chlorine has an A_r value less than 40% of the total mass			2	2		
	(b)	(i)		 diagram showing $\delta+$ and $\delta-$ (1) lone pair on OH^- (1) curly arrows from lone pair **and** to break C—Cl bond (1) propan-1-ol and Cl^- formed (1)	4			4		

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Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
		(ii)		nucleophilic substitution / hydrolysis	1			1		
	(c)	(i)		reflux/heat with ethanolic / alcoholic sodium hydroxide	1			1		
		(ii)		CH_2=$CHCH_2CH_2CH_3$ (1) CH_3CH=$CHCH_2CH_3$ (1) only credit one *E*/*Z* isomer award (1) for **both** correct names if no structures given pent-1- and pent-2-ene		2		2		
				Question 12 total	**9**	**2**	**2**	**13**	**0**	**3**

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